

ANALYSIS INTO RAIN AND CRASH DATA IN MONTGOMERY COUNTY 2025

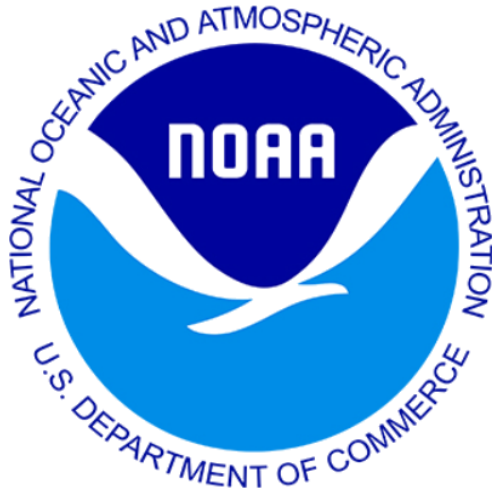
PAUL HANSCOM

WHAT TYPE OF
CAR IS THIS?



DATA

- THE DATA I AM USING FOR WEATHER IS A DATASET I REQUESTED FROM NOAA THE NATIONAL OCEANIC AND ATMOSPHERIC ADMINISTRATION.
- SPECIFICALLY IT CONTAINS THE DAILY RAINFALL RECORDED AT EACH STATION IN MARYLAND
- I CLEANED THE DATA TO ONLY INCLUDE STATIONS IN MONTGOMERY COUNTY



- THE DATASET I AM USING FOR CAR CRASHES, IS A RECORD FROM DATA MONTGOMERY.
- IMPORTANT VARIABLES INCLUDE LATITUDE, LONGITUDE, AND TIME.

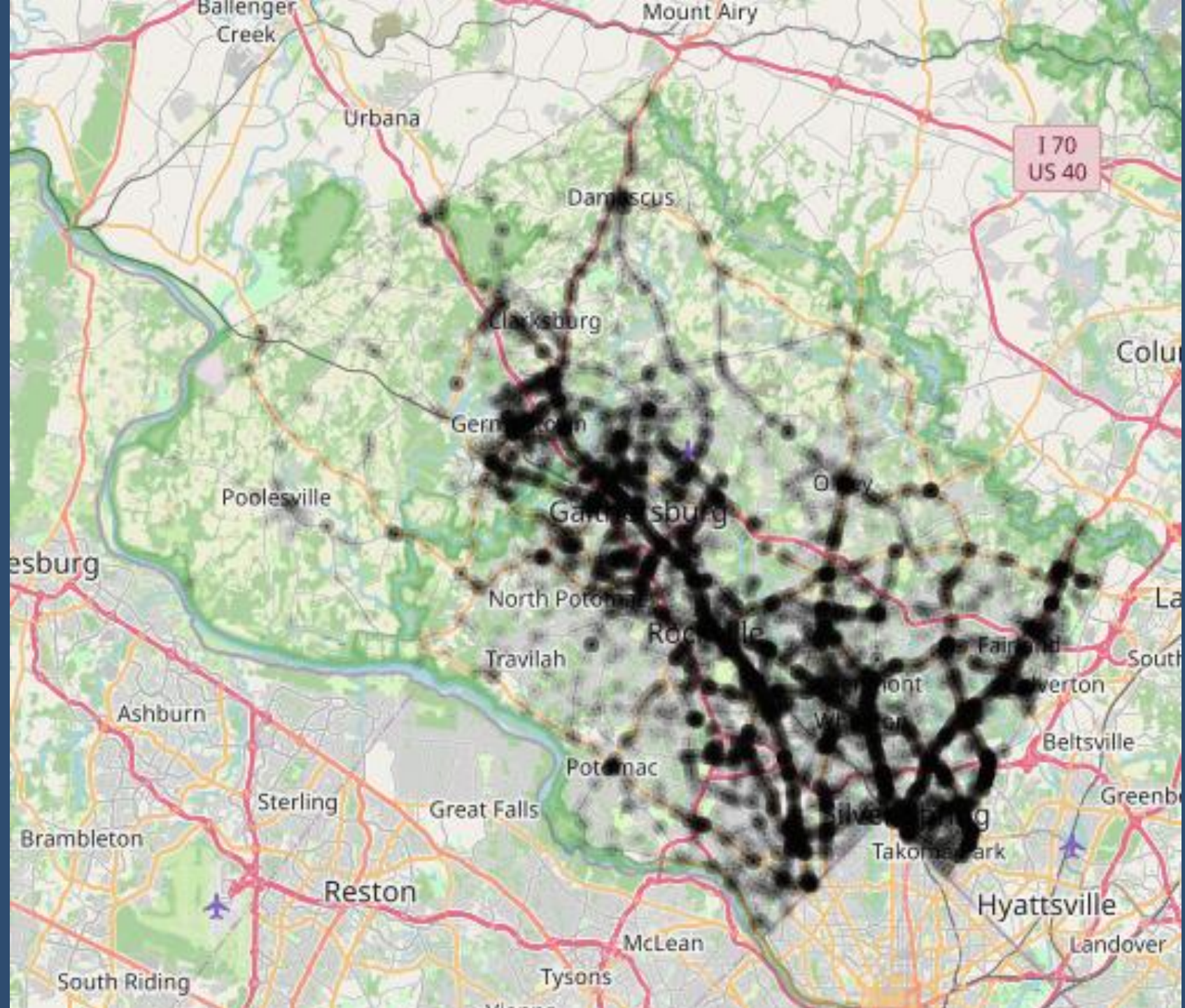


TOOLS

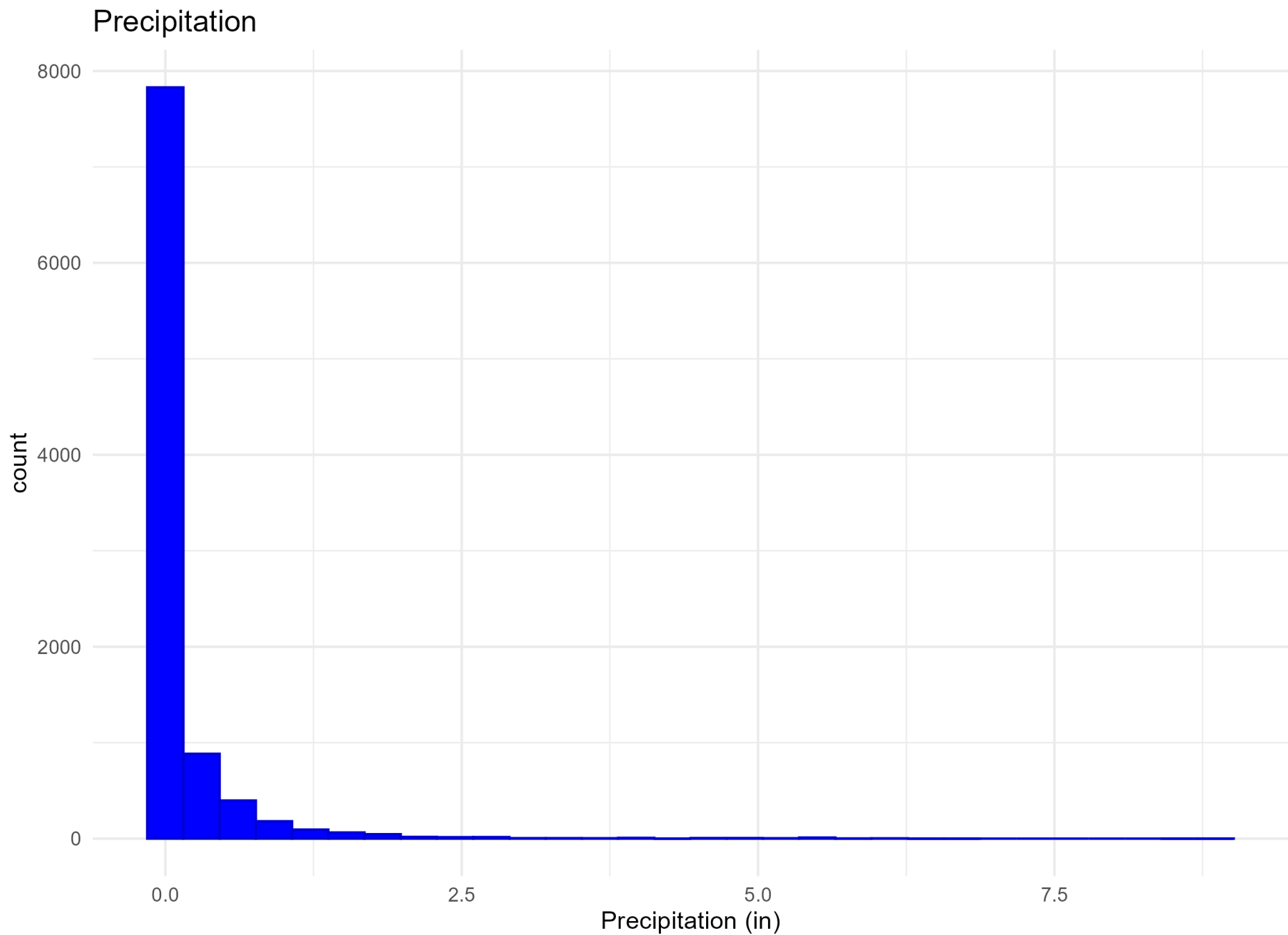
- R STUDIO (INTERFACE)
- LEAFLET (FOR MAPS)
- GSTAT (FOR INTERPOLATION)
- LUBRIDATE (FOR MANIPULATION OF DATE VARIABLES)
- MAPSHOT (FOR CAPTURING IMAGES OF MAPS)
- FFMPEG (FOR COMPILING IMAGES INTO A VIDEO)



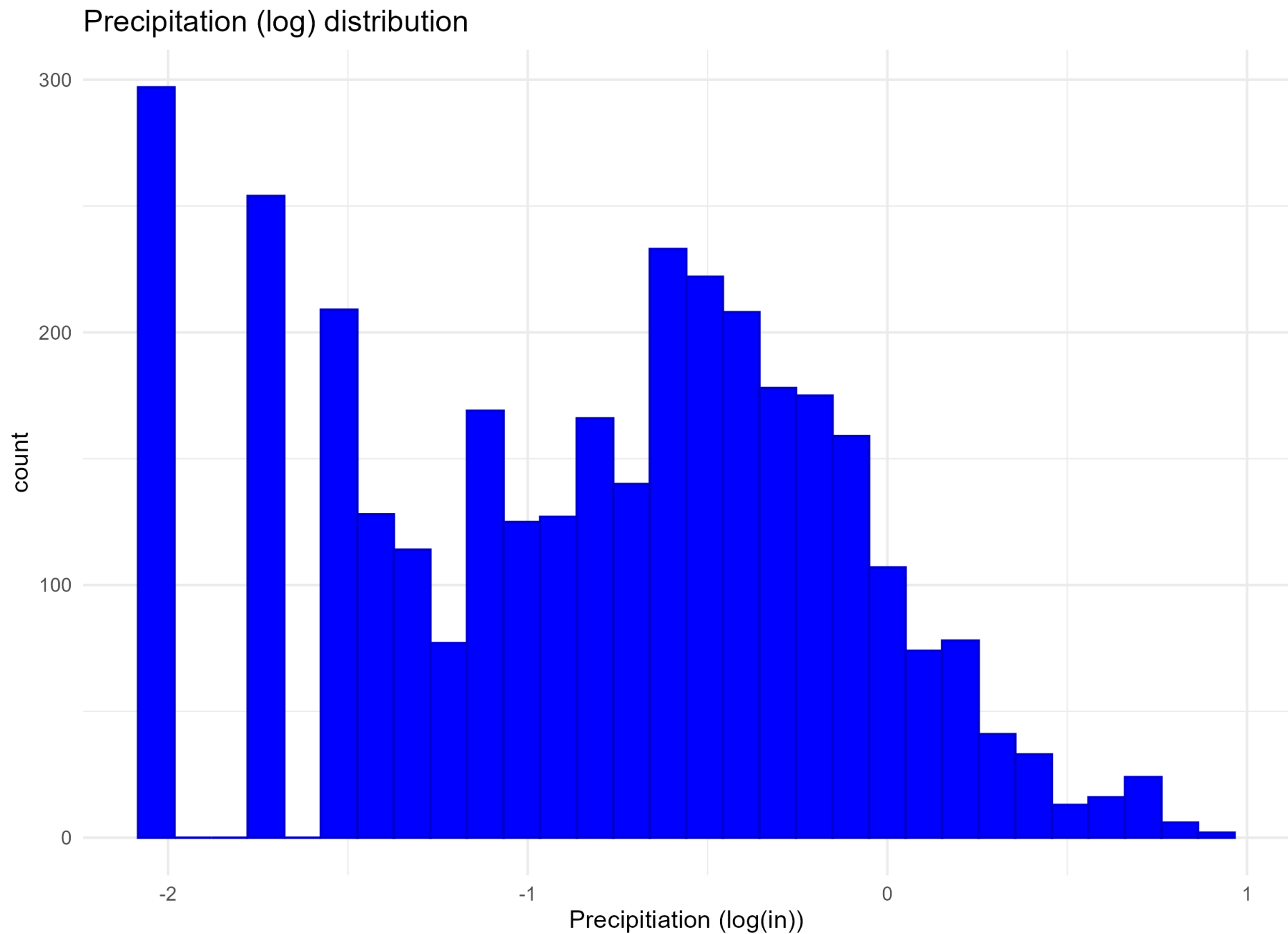
CRASH DATA



WEATHER DATA

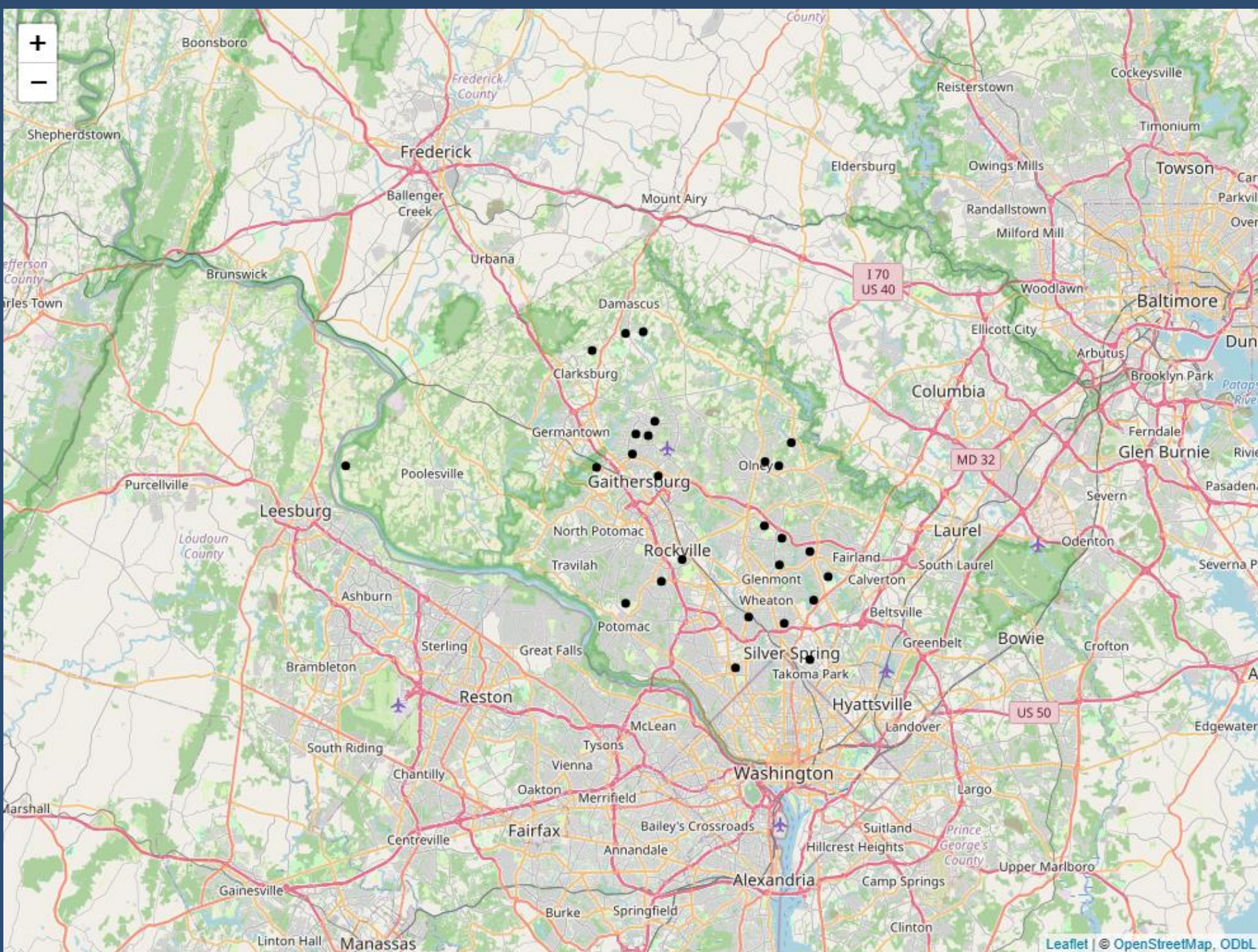


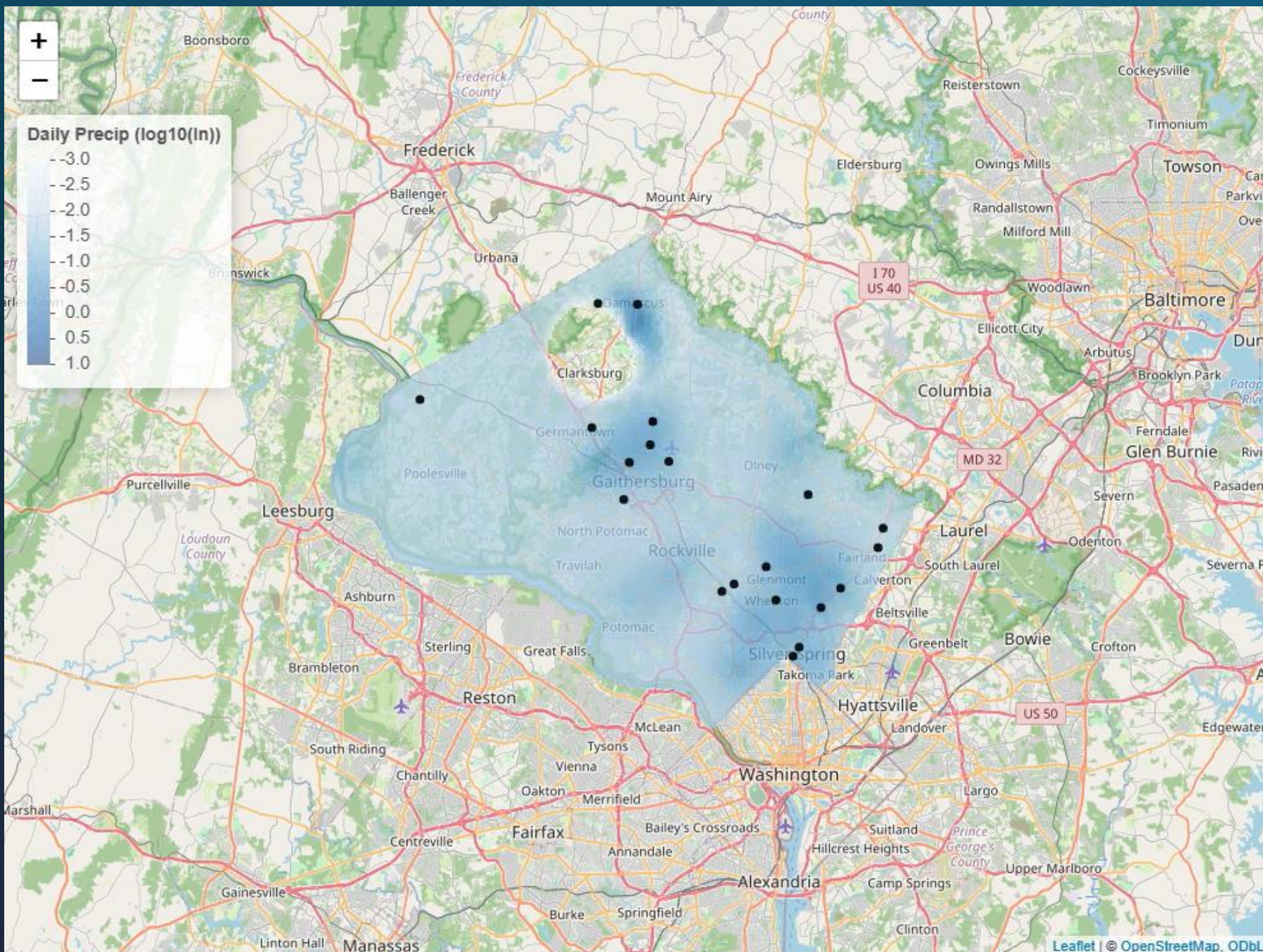
data from NOAA available <https://www.ncei.noaa.gov/cdo-web/>



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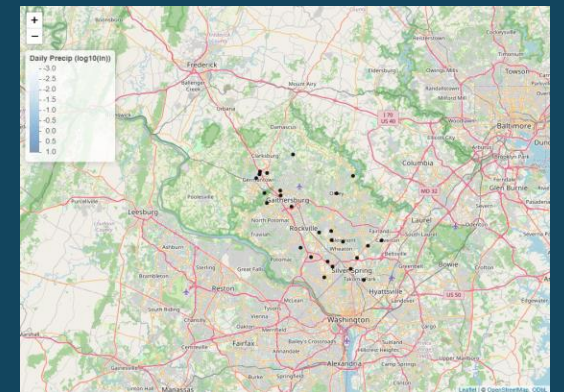
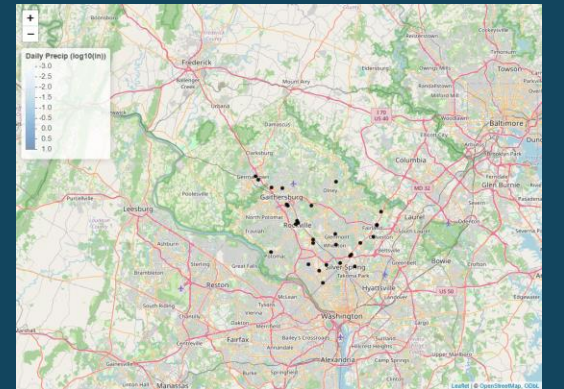
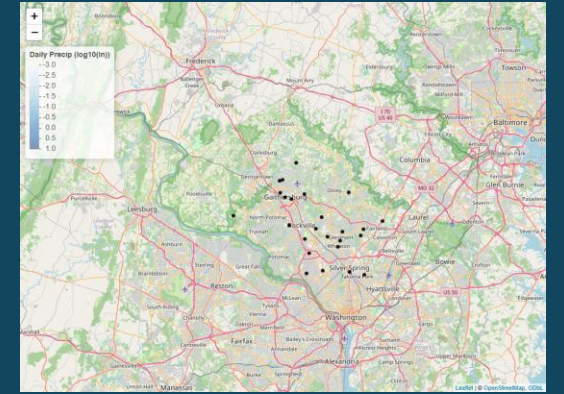
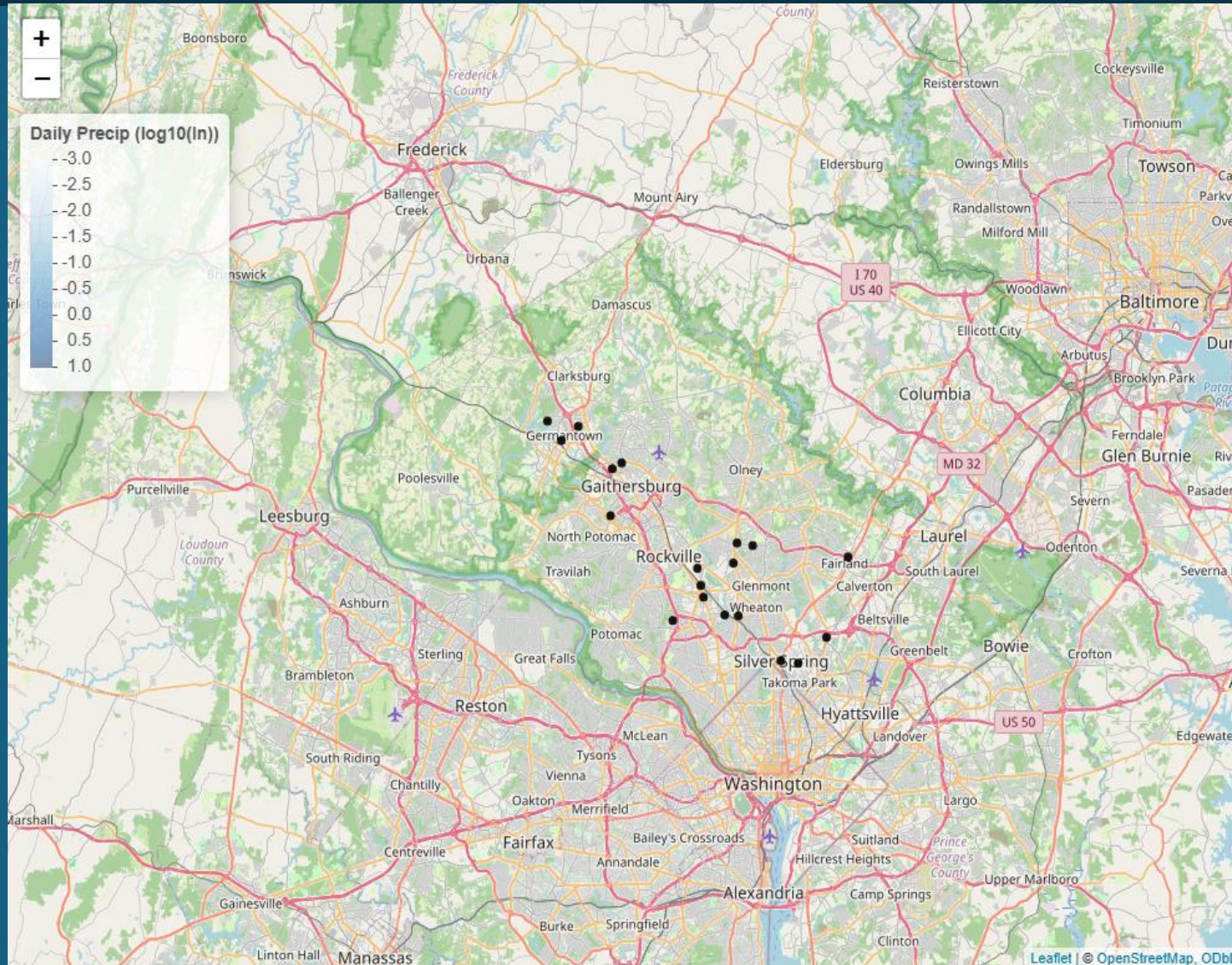
WEATHER DATA (CONT.)

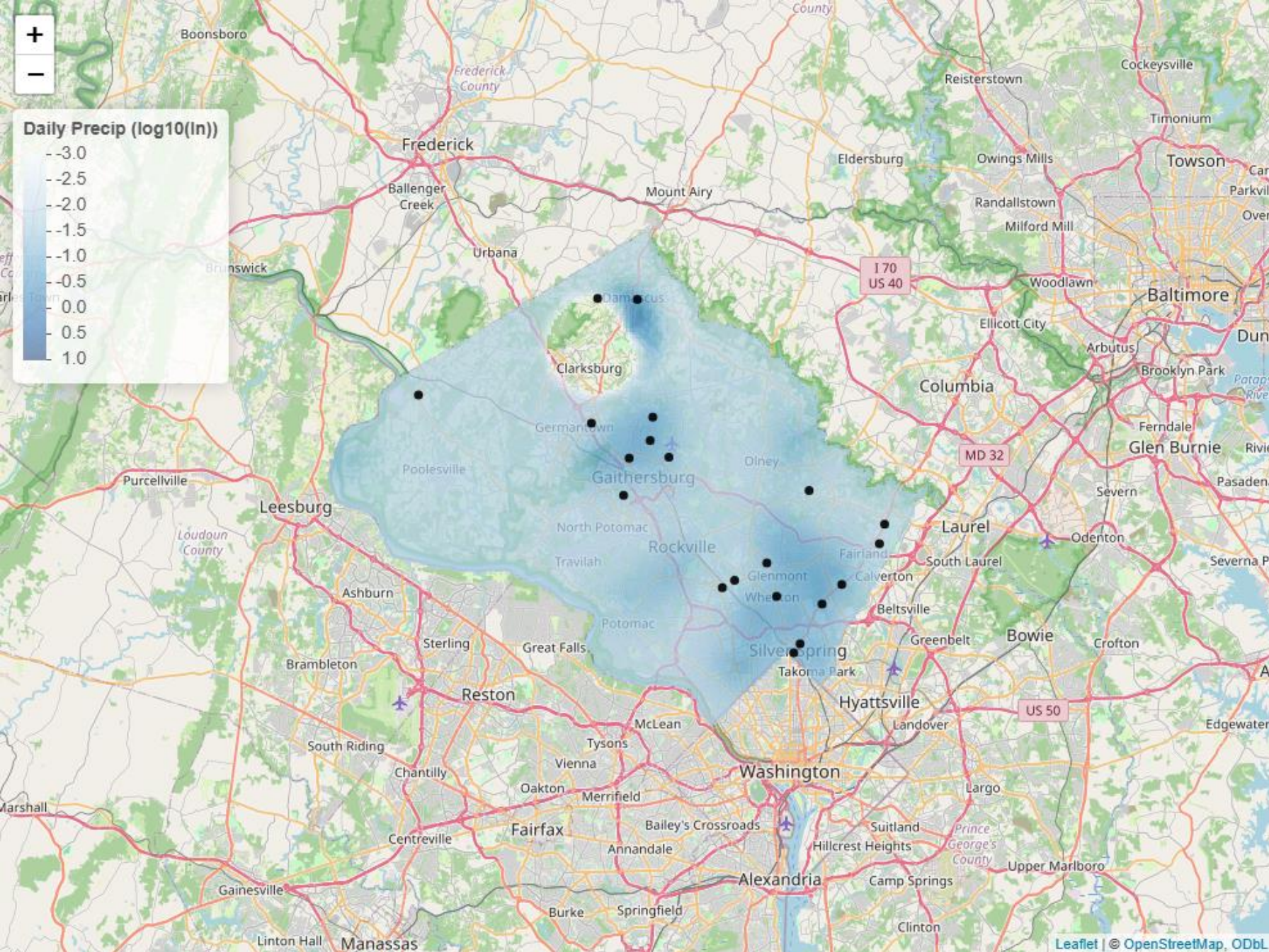




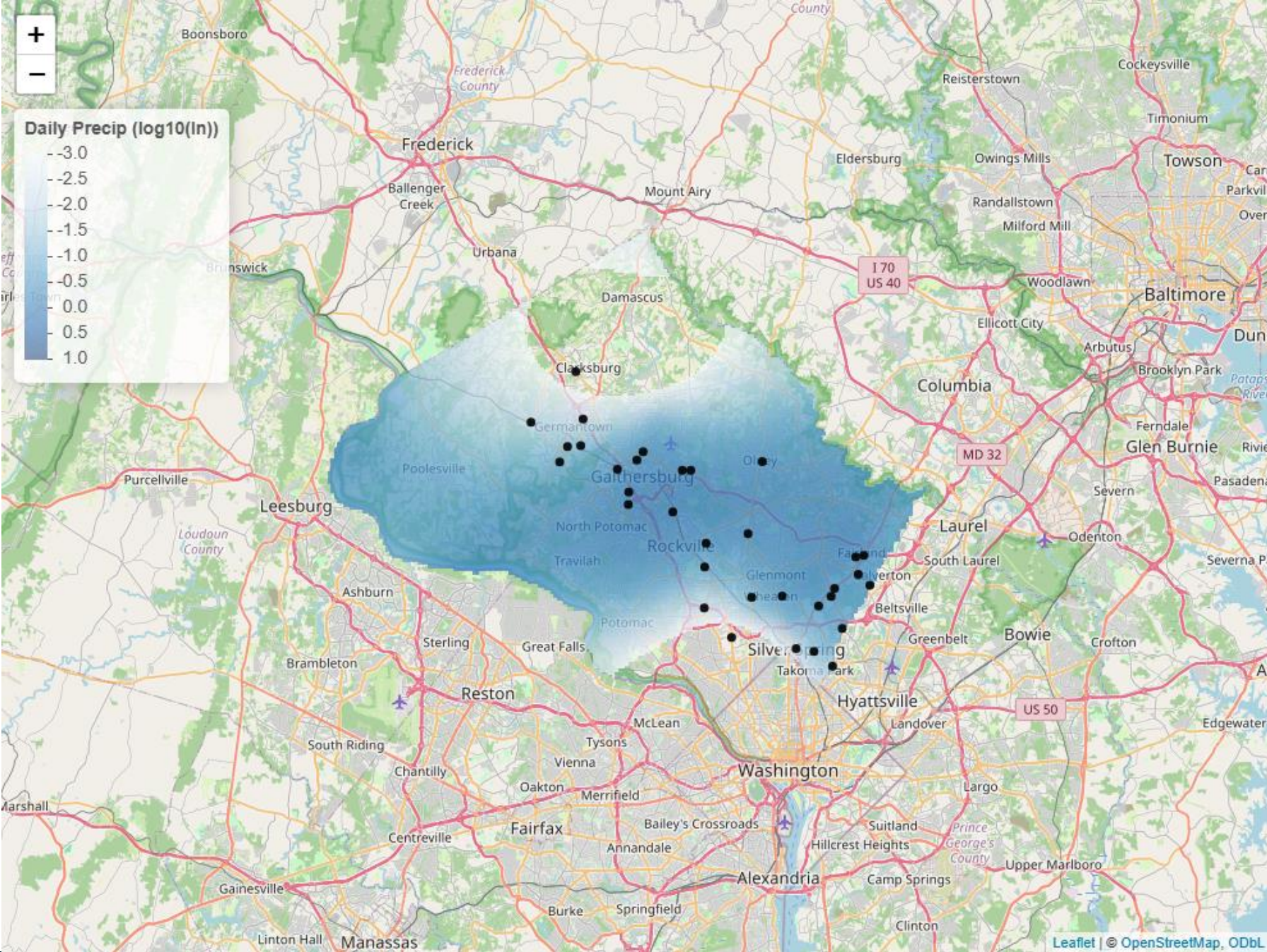
**VARIOGRAMS
MODEL SPATIAL
VARIABILITY WHICH
IS THEN USED TO
WEIGHT KRIGING.**

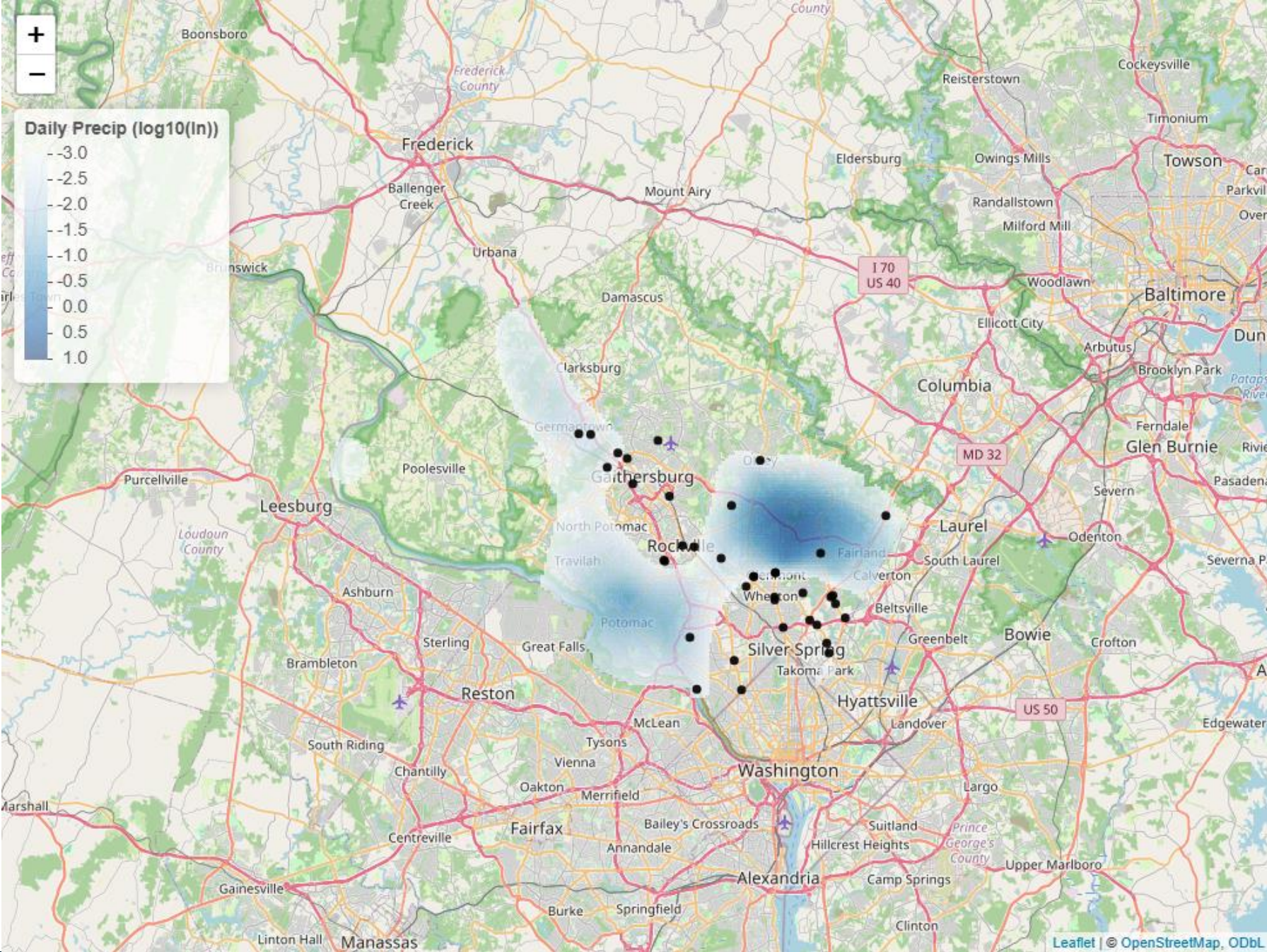
MOST MAPS



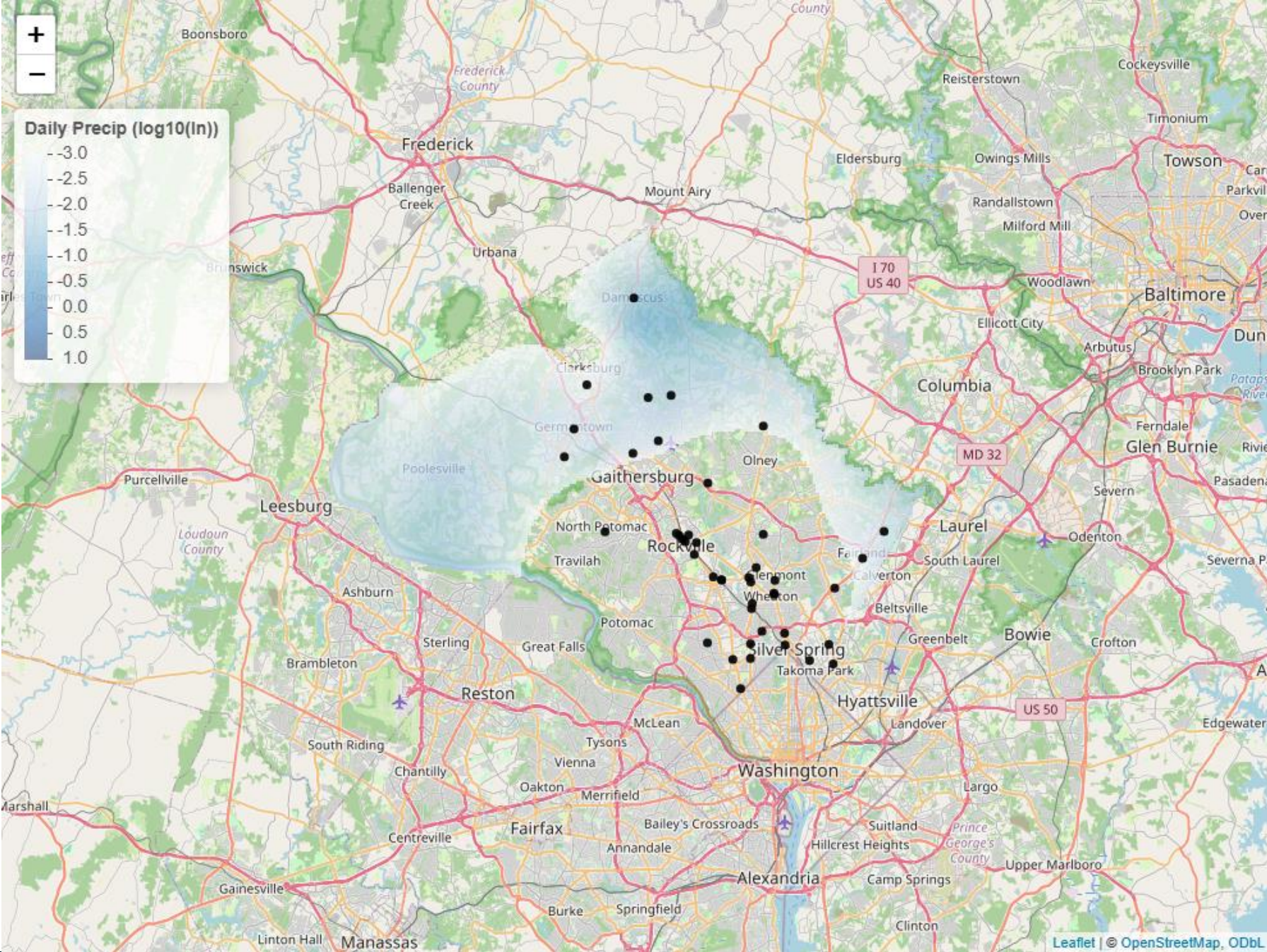


JANUARY 1ST, 2025



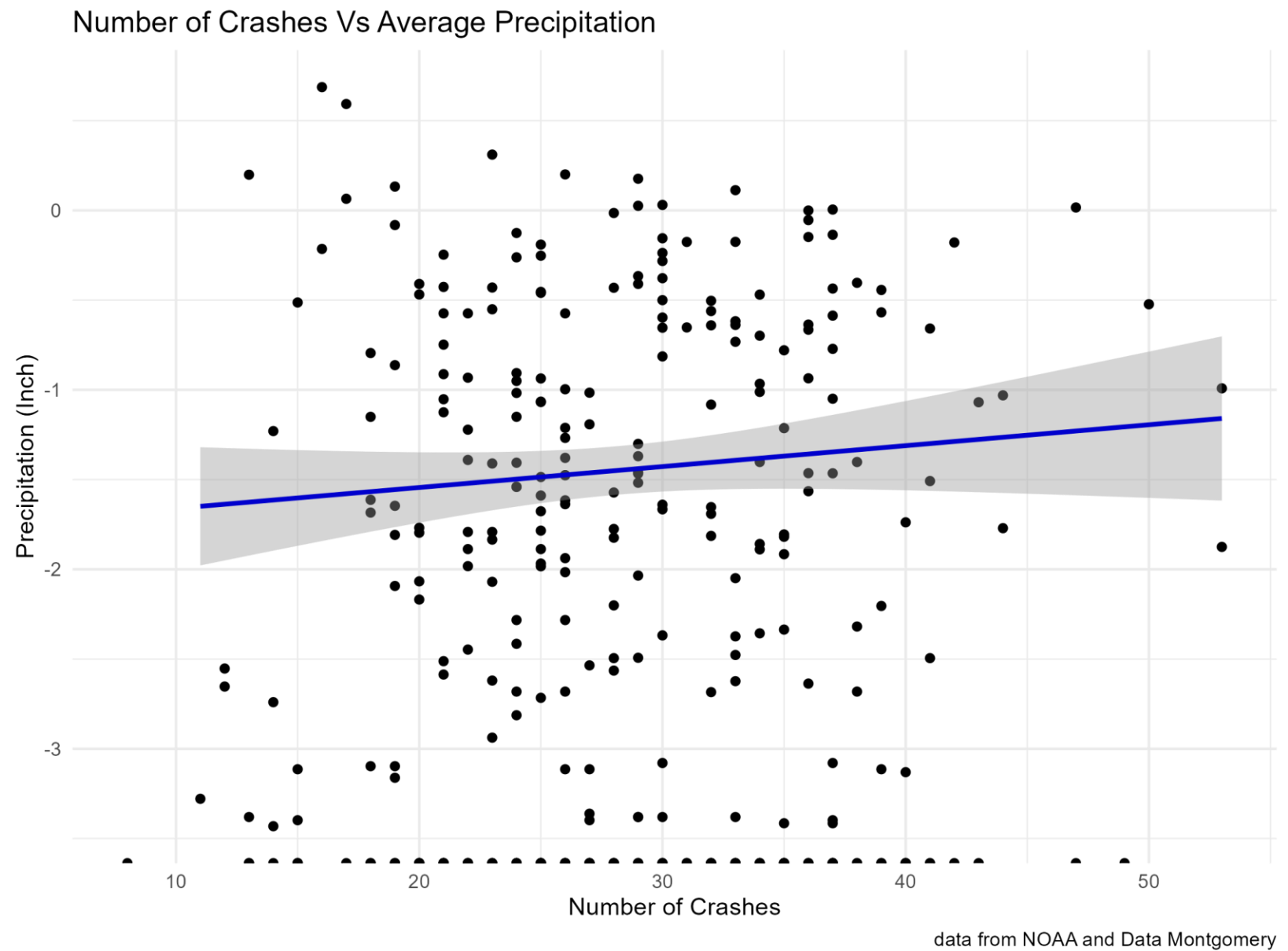


**JUNE
27TH, 2025**



**SEPTEMBER
18TH, 2025**

CORRELATION TEST



P-VALUE = 0.1375, MEANING NOT STATISTICALLY SIGNIFICANT

REFLECTION & NEXT STEPS

- CREATING THE MAPS USING KRIGING WAS CHALLENGING, AS I HAVE NEVER DONE IT BEFORE.
- I REALLY LIKE HOW THE MAPS CAME OUT
- IF I WOULD CHANGE ANYTHING I WOULD EXPAND MY SCOPE I HAVE WEATHER DATA FOR ALL OF MARYLAND AND I THINK IT WOULD BE COOL TO LOOK AT. IM ON THE LOW END FOR DATAPPOINTS FOR KRIGING I HAVE ENOUGH IN THEORY BUT MORE WOULD LIKELY IMPROVE THE MODEL
- AS A FUTURE STEP I WOULD LIKE TO FIND SATELLITE WEATHER DATA FOR A GIVEN DAY TO COMPARE MY KRIGING TO REAL DATA.

ACKNOWLEDGMENTS

- TO MY PAST DATA PROFESSORS, LORI PERINE AND DAESHIK PARK
- TO PROFESSOR ALRAEE FOR HER HELP IN THIS CLASS AND OTHERS
- TO MY FELLOWS IN MY SESSION DATA 205 FOR THEIR FEEDBACK
- TO THE EARLY COLLEGE DATA COHORT
- TO DOCTOR SHROYER FOR THEIR HELP IMPROVING MY WEATHER VISUALIZATIONS.
- TO GENEVIEVE MONTGOMERY FOR DRIVING ME AROUND TO TAKE A PICTURE FOR MY SLIDES
- TO DATA MONTGOMERY AND THE MONTGOMERY COUNTY GOVERNMENT FOR THEIR PARTNERSHIP WITH THIS PROGRAM.
- TO FRIENDS AND FAMILY OUTSIDE OF SCHOOL FOR THERE EMOTIONAL SUPPORT
- AND TO THE PEOPLE WHO'VE BEEN PLAYING DND WITH ME FOR GIVING ME SOMETHING ELSE TO FRET OVER BESIDES THIS PROJECT.